

full servo block

$$G_1(s) = \frac{3.43}{s+2.14375}$$



dominant-pole
approximation

$$G'_1(s) \approx 1.6$$

rewrite the servo model in tail-one form, remove the fast pole, and keep the dominant low-frequency behavior

compare the unit-step responses of the original model and the reduced model around the chosen gain $K = 0.6064$

the two curves are close, so the dominant-pole design is acceptable for the main maneuvering dynamics